TEXORA · LECTURE SHEET

Fibre Diameter

Diameter is the natural measure of fineness for round fibres — above all wool, where a single micrometre moves the price. This sheet covers what diameter means for non-round fibres, and the projection-microscope, airflow and modern image-analysis methods.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 4 · LEVEL 3 TERM 1

1. Diameter, fineness and cross-section

Fibre diameter is the width of the fibre cross-section, expressed in micrometres (µm). For a truly circular fibre (glass, many synthetics) one number suffices. Natural fibres are not circular:

- **Wool** is near-elliptical (ellipticity typically 1.15–1.3 : 1); diameter means the average of the projected width over many fibres and orientations.
- **Cotton** is a collapsed, convoluted ribbon (Topic 5); it has no single diameter, and its fineness is handled through linear density and micronaire instead. Quoting a cotton "diameter" (about 11–22 µm ribbon width) without that warning misleads.
- **Silk** is triangular; flax polygonal. For such fibres **linear density** (Topic 10) is the honest fineness measure, and diameter is an auxiliary quantity.

Diameter matters because it sets, for a given fibre density, the fibre's linear density, its bending stiffness (which rises with the *fourth power* of diameter — a fibre twice as thick is about sixteen times as stiff), the number of fibres in a yarn cross-section, fabric handle and prickle (wool fibres above roughly 30–32 µm trigger skin prickle), and dyeing rate through surface area per unit mass.

$$\text{Cross-sectional area (circular fibre)} \quad A = \pi\, d^2 / 4$$

$$\text{Linear density (tex)} = A\ (\text{cm}^2) \times \text{density}\ (\text{g/cm}^3) \times 10^5$$

WORKED EXAMPLE — FROM DIAMETER TO LINEAR DENSITY

A wool fibre of mean diameter 20 µm and density 1.30 g/cm³. Radius = 10 µm = 0.0010 cm; $A = \pi \times (0.0010)^2 = 3.14 \times 10^{-6}$ cm². Mass per cm = $3.14 \times 10^{-6} \times 1.30 = 4.08 \times 10^{-6}$ g/cm. Per kilometre (10^5 cm): **0.41 tex** (4.1 dtex). Doubling the diameter to 40 µm would quadruple the area and the tex — the reason a few coarse fibres dominate a blend's stiffness and handle.

2. The reference method — projection microscope

1. Snippets of fibre about 0.4–0.8 mm long are cut (microtome-like snippet cutter) from a conditioned, randomly drawn sample — snippetting removes length-bias and presents every fibre equally.
2. Snippets are mounted in a non-swelling medium (e.g. liquid paraffin / cedarwood-type mounting oil as the standard directs) under a cover glass.
3. The slide is projected at a standard magnification (×500) onto a screen; the projected width of each snippet is measured against a graduated scale / wedge. Rotating the stage is unnecessary because the snippets lie at random orientations, so elliptical fibres are sampled fairly.
4. At least 600 fibres are measured for a sale-grade result (more where the diameter CV is high). The **mean fibre diameter** and its **standard deviation and CV%** are reported — for wool, typical within-lot CV is 20–25 %, so the distribution, especially the coarse tail, matters as much as the mean.

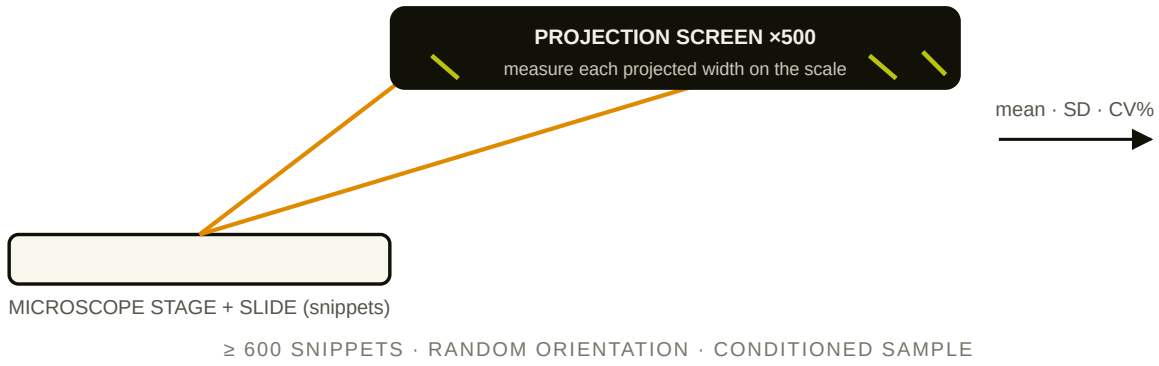


Figure 1 — Projection microscope principle. Snippetting makes the measurement a diameter survey of the whole sample instead of a study of a few long fibres.

3. Faster and modern methods

METHOD	PRINCIPLE	NOTES
Airflow (wool)	Air is drawn through a constant-mass, constant-volume plug of fibre; finer fibre = more surface = greater resistance = lower flow. Calibrated against projection-microscope reference wools.	Rapid mill / topmaker method; gives mean diameter only, no distribution; sensitive to plug preparation.
OFDA (Optical Fibre Diameter Analyser)	Snippets spread on a slide are scanned automatically; image analysis measures each snippet's width.	Thousands of fibres in minutes; mean, SD, CV, coarse-fibre (>30 µm) percentage and curvature.

Laserscan	Snippets in a flow cell cross a laser beam; each fibre's width is measured as it passes.	The IWTO-reference modern instrument for wool lots; full distribution.
Sirolan / image methods for other fibres	Microscope + camera + software on prepared cross-sections or longitudinal mounts.	For synthetics, diameter follows from the filament's specified denier and density as a check.

For **man-made filaments**, diameter is usually controlled through linear density at manufacture and verified by weighing (Topic 10); direct diameter measurement is used for monofilaments, optical fibres and technical filaments where the dimension itself is the specification.

4. Interpretation — wool as the model case

- **Spinning limit.** A yarn cross-section needs a minimum number of fibres (roughly 40–60 for worsted spinning) to spin evenly; mean diameter therefore caps the finest count: finer wool (18–20 µm) spins fine counts, coarse wool (30 µm+) is confined to coarse counts, carpets and blankets.
- **Comfort factor.** The percentage of fibres ≤ 30 µm; high comfort factor = low prickle for next-to-skin garments.
- **Diameter profile along the staple.** Wool diameter varies along the fibre with the sheep's nutrition and season; tender (thin-waisted) wool breaks in processing — a diameter *distribution along length* problem that a single mean hides.
- **Blending.** Lots are blended to a target mean diameter; because stiffness rises as d^4, a small proportion of coarse fibre ("kemps", medullated fibres) spoils handle out of proportion to its mass.

WORKED EXAMPLE — MEAN DIAMETER AND ITS CONFIDENCE

600 snippets give mean diameter 22.0 µm with SD 5.0 µm (CV = 5.0/22.0 × 100 = **22.7 %**). Standard error of the mean = 5.0 / √600 = **0.20 µm**. So the lot mean is known to about ±0.4 µm at 95 % confidence (mean ± 2 SE) — tight enough for pricing — while individual fibres still range widely, which is why the coarse-fibre percentage must accompany the mean on any next-to-skin specification.

5. Precautions

- Measure conditioned samples; wool swells across the diameter as regain rises, so an unconditioned sample reads coarse on a humid day.
- Mounting medium must not swell or shrink the fibre; fibres must not overlap on the slide (overlaps read as one coarse fibre).

- Calibrate magnification with a stage micrometer at the session's start; a 2 % magnification error is a 2 % diameter error on every fibre.
- Cut snippets from several zones of the lot (Topic 1 sampling) — diameter varies between body regions of the fleece and between bales.

Glossary

Mean fibre diameter — average projected width of fibre snippets, in µm. **Ellipticity** — ratio of major to minor axes of the cross-section. **Snippet** — a sub-millimetre cut length of fibre prepared for diameter measurement. **Comfort factor** — percentage of fibres of diameter ≤ 30 µm. **Medullated fibre / kemp** — a fibre with a hollow or partly hollow core; coarse, stiff and dye-resistant. **OFDA / Laserscan** — automated image / laser instruments measuring thousands of snippets.

Key points

• Diameter (µm) is the fineness measure for near-round fibres; for ribbon-like cotton, linear density replaces it.
• Stiffness rises as d^4 — coarse fibres dominate handle and prickle far beyond their mass share.
• Reference method: projection microscope at ×500 on ≥600 snippets; report mean, SD, CV and coarse %.
• Airflow gives a rapid mean only; OFDA and Laserscan give full distributions in minutes.
• Condition the sample and calibrate magnification — diameter is humidity- and scale-sensitive.

Practice

1. A circular polyester filament has diameter 15 µm and density 1.38 g/cm³. Calculate its linear density in dtex.
2. Two wool lots both average 21 µm; lot A has CV 19 %, lot B CV 27 %. Which is the safer choice for a next-to-skin knit, and why?
3. Why are snippets, not whole fibres, measured in the projection microscope method?
4. A laboratory forgets to condition wool samples during a humid week. What happens to the reported mean diameter, and why?

Answer key

1. $A = \pi \times (7.5 \times 10^{-4}\ \text{cm})^2 = \pi \times 5.625 \times 10^{-7} = 1.767 \times 10^{-6}\ \text{cm}^2$. Mass/km $= 1.767 \times 10^{-6} \times 1.38 \times 10^5 = 0.244$ tex = **2.44 dtex**.
2. Lot A. The prickle and stiffness come from the coarse tail; at equal mean, the lower-CV lot has far fewer fibres above ~30 µm, so handle is softer and spinning more even.

3. Snippets randomise orientation (elliptical fibres are sampled fairly), remove selection bias toward easily-gripped long fibres, and let hundreds of fibres be mounted and measured on one slide.
4. Reported diameter rises: wool absorbs moisture and swells mainly across its cross-section, so each projected width is genuinely larger at high regain. The error is systematic until conditioning is restored.

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